



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	70.0	0.0	0.0	°C
p1	10.0	0.0	0.0	bar(a)
☐ p2	7.0			bar(a)
☒ Δp	3.0	0.0	0.0	bar
Cv	330.07			GPM(US)
☐ qm	489,080.0			kg/h
☒ qv	500.0	0.0	0.0	m³/h
LpAe	80.1			dB(A)
Non-critical				

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	978.16			kg/m³
ρ2	978.0			kg/m³
☐ η1	0.40412			mPa s
☒ ν1	0.41314			mm²/s
pv1	0.31201			bar(a)
tv1	179.88	373.95	373.95	°C
cF1	1,556.5			m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

STD



Calculation Sheet

Valve configuration

Valve type	<i>Straight globe valve</i>
Trim type	<i>Contoured plug</i>
Flow direction	<i>FTC</i>
Valve performance class	<i>Multi stage valve</i>
Protection	<i>Non-hardened</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Linear</i>	
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class	<i>class 600</i>	
Maximum operating pressure	p,max	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.802</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.719</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.472</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
u2	! <i>68.537</i>			m/s
LpAe	<i>80.1</i>			dB(A)
Ri	! <i>1.7</i>			-

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>1,556.5</i>	m/s
Pressure level	LA1	<i>165.0</i>	dB
Sound pressure level	Lpi	<i>165.0</i>	dB
Transmission loss	TL	<i>-72.5</i>	dB





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Noise prediction data (continued)

Sound power level (A-weighted)	LwAe	90.2	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	80.1	dB(A)

Comments:

Reliability index - Ri

High velocity: 100 %
Recommended solution: Standard valve

Legend

- ✖ Errors
- ! Alarm

